

Assignment-1

Digital Electronic (PCCEEE402)

1. What is a binary variable?
2. What is Boolean algebra?
3. State De Morgan's theorem.
4. Prove that: $ABCD + AB(CD)' + (AB)'CD = AB + CD$
5. Prove that: $AB + (AC)' + A B'C(AB + C) = 1$
6. Reduce the following expressions k- map:
 - a. $F1 = m1 + m2 + m3 + m5 + m7 + d(m0 + Dm6)$
 - b. $F2 = m(0, 1, 4, 7, 13, 14) + d(5, 8, 15)$